

# Micropayments and user behavior on digital platforms: Evidence from a Bitcoin-based discussion forum\*

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May 11, 2026

## Abstract

We study how financial micro-incentives shape user behavior on a digital platform that uses Bitcoin micropayments. On the platform, users must pay a small fee to post content, and can earn peer-to-peer tips from other users who value their contributions. Using variation in posting fees across the platform's sub-forums and over time, we estimate the elasticity of content quantity and content quality with respect to posting cost. We find that doubling the posting fee reduces the number of posts by 16.8% but increases the quality of posts by 16.4%, consistent with a signaling model in which posting costs screen for higher-quality contributions. We also find evidence consistent with users learning over time which types of content are valued by others, leading to gradual improvements in content quality. Finally, we show that users who are unable to recoup their posting costs through tips are more likely to leave the platform, while profitable users are more likely to stay. These findings provide empirical evidence on the behavioral effects of micropayment and peer-to-peer incentives in digital platforms.

*Keywords:* micropayments, digital platforms, user-generated content, Bitcoin, Lightning Network, market design, signaling

*JEL Classification:* D82, D83, L82, L86

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\*The views expressed in this paper are solely those of the authors and do not necessarily reflect the views of Stacker News. Stacker News provided the data, but did not exercise editorial control over the content of this paper, nor over the analysis, interpretation, or reporting of the results. All errors are those of the authors. Corresponding author: Edward Kung, Department of Economics, California State University-Northridge, [edward.kung@csun.edu](mailto:edward.kung@csun.edu). All computer code used for the analysis is open source and available at <https://github.com/ed-kung/sn-research>.

# 1 Introduction

The design of incentive structures on digital platforms is a central question in the economics of user-generated content (Kraut et al., 2012; Luca, 2015; Aridor et al., 2024). Most platforms that host user-generated content rely on direct incentive mechanisms—such as advertising revenue sharing, algorithmic visibility, and reputation systems—to encourage content production and discourage low-quality contributions. Direct financial mechanisms, such as requiring users to pay a fee to post content, or allowing peer-to-peer payments between users, remain rare. This is partly due to frictions in the payments infrastructure: the value of any individual post or comment is typically too small to justify credit card transaction fees, and international platforms face additional frictions related to currency exchange and cross-border banking regulations. However, the emergence of low-cost digital payment rails, such as Bitcoin’s Lightning Network, has made it technically feasible for platforms to integrate micropayments into their core design, opening the door to new incentive structures that were previously impractical.<sup>1</sup>

There is good reason to believe that micropayment mechanisms could meaningfully influence user behavior on digital platforms. Signaling theory suggests that posting costs can serve as a screening device, deterring low-quality contributions while allowing users with high-quality content to signal their type by paying the cost (Joseph and Thevaranjan, 2008). Empirically, Tchernichovski et al. (2019) showed that introducing a small fee to an online rating system filtered out low information signals and increased the correlation between review ratings and quality. Other empirical studies

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<sup>1</sup>The Lightning Network is a decentralized payments network that is built on top of Bitcoin. It enables faster and lower cost Bitcoin transactions than the base layer would allow. It works by letting users maintain peer-to-peer payment channels that allow them to “float” Bitcoin between each other indefinitely, until final settlement occurs on chain. Users can pay others with whom they have no direct channel by routing payments through other channels, usually at a percentage fee.

have found that financial incentives in the form of revenue sharing can influence the quantity and quality of user-generated content (Sun and Zhu, 2013; El-Komboz et al., 2023; Kerkhof, 2024). However, none of these studies have investigated the behavioral effects of posting fees and direct peer-to-peer micropayments in an operational platform setting.

This paper provides empirical evidence on how micropayment incentives shape user behavior, using data from a Bitcoin-based discussion platform called Stacker News. Stacker News is an internet discussion forum—similar in format to Hacker News or Reddit—that integrates Bitcoin micropayments into its core design through the Lightning Network. Users must pay a fee denominated in Bitcoin (usually on the order of 1-100 satoshis) to post content in any of the platform’s topic-specific sub-forums, called territories.<sup>2</sup> Users can also send each other peer-to-peer Bitcoin micropayments, called “zaps”, to reward each other for content that they value. The combination of posting fees and peer-to-peer tipping creates a setting in which both the cost and reward sides of content production are mediated by real financial transactions, making Stacker News a natural field setting for studying how micropayment incentives influence user behavior.

We exploit variation in posting fees across territories and over time to estimate the effect of posting costs on the quantity and quality of user-generated content. Using a difference-in-differences framework with territory and time fixed effects, we find that doubling the posting cost in a territory reduces the number of posts by 16.8%, but increases the quality of posts—measured by zaps received from other users—by 16.4%. Event study analysis confirms that these effects follow fee changes rather than preceding them. Beyond the posting cost mechanism, we also find evidence that users

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<sup>2</sup>A satoshi (or sat) is 1/100,000,000th of a Bitcoin. At the time the research was conducted, 1,000 sats was worth about USD \$1.

learn what types of content other users value over time and that this shapes their posting behavior. Specifically, we find that users whose previous high-quality posts (measured along objective dimensions like word count and images) received more tips than expected are subsequently more likely to produce similar high-quality content, consistent with a model in which users learn about the returns to effort from the financial feedback they receive. Finally, we find that users who are unable to earn enough tips to cover their posting costs are more likely to leave the platform, while profitable users are more likely to stay—a selection mechanism that could further improve content quality over time.

Our findings contribute to the literature on platform design and user-generated content in several ways. First, we provide some of the first estimates of the price elasticity of user-generated content with respect to posting costs, a parameter that is directly relevant to platforms considering micropayment mechanisms. Second, we provide evidence that peer-to-peer tipping creates incentives that lead to quality improvements over time, extending the literature on financial incentives in content production beyond the revenue-sharing mechanisms studied previously. Third, we document a profitability-driven selection mechanism through which micropayments affect not only the behavior of individual users, but also the composition of the user base over time. While our setting is a single platform with a specific user community organized around Bitcoin, the underlying mechanisms we identify—cost-based screening, tip-driven learning, and profitability-based selection—are general features of micropayment systems and are likely to operate on other platforms that adopt similar designs. Our results may offer practical guidance for platform designers considering the integration of micropayment infrastructure into user-generated content platforms.

## 1.1 Related literature

Our paper relates to a substantial literature on the economics of platform design and user-generated content (UGC). Luca (2015) surveys the economics of UGC, documenting how quality is shaped by promotional content, peer effects, contributor biases, and self-selection into the decision to contribute, noting that non-pecuniary incentives are often used to steer contributions. Ghosh and McAfee (2011) provide a game-theoretic model of UGC incentives in which contributors are strategic and motivated by exposure, showing that when exposure is independent of quality, platforms will be flooded with low-quality content. This paper speaks to their findings by showing what happens when quality is financially incentivized and when contributions are screened by an entry fee. Ghosh and Hummel (2014) extend this framework to rank-order mechanisms, analyzing how platforms can use ranking systems to incentivize effort, which is relevant to how Stacker News uses zap-based rankings to surface content. Liu and Feng (2021) develop a model showing that monetary incentives can either increase or decrease UGC contributions depending on heterogeneity in contributor motivation. Our work provides an empirical test of the net effect of direct monetary incentives, plus additional findings regarding the effect of entry fees.

On the empirical side, Muchnik et al. (2013) demonstrate in a large-scale randomized experiment that prior positive social signals create significant herding effects on subsequent ratings, raising important questions about whether peer-generated evaluations reflect true quality or social influence. Our use of zaps as a quality measure is subject to similar concerns, though the fact that zaps on Stacker News require real monetary outlays may mitigate the bias relative to costless upvote systems. Chen et al. (2010) conduct a field experiment on MovieLens showing that social comparisons (informing users about how their contribution levels compare to other users)

significantly increases contributions. Our finding that users learn from their personal zap history (Section 5) extends this insight by showing that financial feedback can also drive higher effort, in addition to social feedback. Relatedly, Paridar et al. (2025) study online tipping on a UGC platform and show that tipping behavior is shaped by evolving social norms and information disclosure. Our contribution differs in that we study a platform where tipping is combined with mandatory posting costs, allowing us to examine both the reward side and the cost side of micropayment incentives.

The design of digital payments systems is also relevant to our setting. Au and Kauffman (2008) analyze the economics of mobile payment adoption, identifying transaction costs, network effects, and stakeholder incentive alignment as key determinants of whether new payment technologies achieve viability. The emergence of Bitcoin’s Lightning Network has substantially reduced the transaction cost barriers they identified, making the kind of micropayment-mediated platform we study technically feasible at scale. Kraut et al. (2012) synthesize social science research on online community design, offering theory-grounded design principles for encouraging contributions and regulating behavior. Their framework highlights the role of financial and reputational incentives in sustaining communities, but they note that direct financial mechanisms remain largely untested in practice—a gap our study helps fill. Lastly, Treiblmaier and Sillaber (2021) provide a list of salient research topics for how blockchain can impact e-commerce. Our findings speak directly to their suggested topics O2 (“how does blockchain impact business models of e-commerce companies”), O3 (“how does blockchain impact organizational structures and value networks of e-commerce companies”), and C4 (“how does blockchain impact segmentation, personalization, and customization in e-commerce”).

## 2 Incentive structures on Stacker News

From its outset, Stacker News was founded with the idea of using Bitcoin micropayments to incentivize quality content and good behavior. Its motto is “Stacker News is trying to fix online communities with economics.” This makes it a natural field setting for studying how micropayment incentives shape user behavior. There are a number of incentive structures built into the design of Stacker News.<sup>3</sup>

**Pay to post.** First, users must pay to post. The posting cost is denominated in sats and varies by territory. A territory is a sub-forum on Stacker News, usually topic-specific. As of October 2025, the most popular territories are `bitcoin`, which focuses on discussions about Bitcoin; `econ`, which focuses on news and discussion about economics; `Stacker_Sports`, which focuses on sports discussion; `AI`, focused on news and discussion about artificial intelligence; and `Politics_And_Law`, focused primarily on politics and current events.

Territories are created and operated by individual users on Stacker News. To create a territory, a user chooses a name for the territory, writes a description, and pays a monthly fee to Stacker News to maintain the territory. As of October 2025, the monthly fee to operate a territory is 50,000 sats.<sup>4</sup> If a territory operator fails to pay the monthly cost, the territory becomes archived. Users are still able to see posts in an archived territory, but they can no longer post in it. Any user (it doesn’t have to be the territory founder) can unarchive a territory simply by choosing to pay the monthly fee, in which case they become the new operator.

Once a new territory is founded, users can post in that territory by paying a posting fee which is set by the territory operator. The territory operator earns 70%

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<sup>3</sup>The following description of how Stacker News works is accurate up to October 5, 2025.

<sup>4</sup>Territory operators can also pay for 12 months at a time for 500,000 sats, or pay for a territory in perpetuity if they pay 3 million sats.

of the posting fee as revenue, and the other 30% goes to a daily rewards pool (more on that later). Territory operators are therefore incentivized to choose a posting fee that optimizes between post quantity, post quality, and revenue earned. If the posting fee is too low, the territory may not earn enough revenue to cover costs, and it may also attract low quality posts. On the other hand, if the fee is too high, users may be deterred from posting in the territory. Users are incentivized to post in the territory that best fits the topic of their post so as to maximize engagement and sats earned net of posting fees. Territory operators can additionally attract users to post in their territories by creating engaging content or by adding value to users' posts.

**Value for value.** Second, users can tip each other directly for content that they like. In the Bitcoin community, systems that allow users to directly tip each other are called “value-for-value” systems. On Stacker News, these tips are called “zaps”. To zap an item, the user clicks a lightning bolt icon which is featured prominently on every post and comment. When the user clicks the lightning bolt, sats are sent from the user to the creator of the item. The default zap amount is 100 sats (about 10 cents at the time of writing), but users can customize their own defaults. They can also long-press the lightning bolt icon to zap a custom amount.

Like Reddit upvotes, zaps are used by the platform to surface high quality content—the default view on Stacker News ranks posts based on the amount of zaps earned by the post, modified by a time decay factor to keep the view fresh. Unlike Reddit upvotes, zaps represent real monetary value transferred from the zapper to the creator of the post. The zapping system incentivizes high quality content in two ways: not only does the system surface higher quality posts to the top of the home page, but it also lets creators of high quality content get paid directly by other users for their work.



While it is clear how zapping incentivizes content creators, it is not as clear what incentives users have to zap. Economists who have studied tipping in non-digital contexts have argued that tipping is not well explained by considerations of future value (Azar, 2020) and have instead focused on altruistic “warm glow” effects (Andreoni, 1990; Crumpler and Grossman, 2008). However, these are contexts where interactions are generally not repeated and where the payment includes a deterministic price along with an optional tip. On Stacker News, users have repeated interactions with each other and there is no fixed reward for content creators other than tips. The incentive model may therefore be closer to that of a “pay-what-you-want” system, in which tipping is rationally used to ensure that producers do not switch to a less advantageous pricing model for consumers (Mak et al., 2015). In the case of Stacker News, the consequence of not tipping could be a reduction in content production that the user enjoys. Whether users zap due to an altruistic desire to reward content that they enjoy, or by a rational incentive to support the continued production of such content, or by a desire to influence the ranking of posts, we do observe a healthy amount of zapping on Stacker News. We also find that the amount of zapping is correlated to objective measures of post quality.

Two additional features of zapping are worth mentioning. First, only 70% of the zap amount is received by the content creator. The remaining 30% is “taxed” and redistributed. 21% of the zap amount goes to the owner of the territory where the item was posted. Through this, territory owners are additionally incentivized to encourage high quality content in their territories. The remaining 9% of the zap goes to the daily rewards pool. This 30% “tax” on zaps is a form of sybil resistance: it discourages users from creating sock puppet accounts to zap their own content.

**Daily rewards.** Third, Stacker News incentivizes user behavior with a daily rewards system. As already mentioned, 30% of posting fees and 9% of zaps go to the daily rewards pool. Users are allocated sats from the daily rewards pool according to a formula that depends on user behavior in that day. Users are rewarded for highly zapped posts and comments, as well as for zapping other highly zapped posts and comments early. The reward system amplifies the already present incentives to post good content, but it also incentivizes users to help identify and surface good content by zapping good content early.

**Market-based content moderation.** Lastly, Stacker News does not employ any content moderators, nor are there formal moderators in any territory other than the territory owners. Instead, Stacker News utilizes a form of market-based content moderation called “downzaps”. A downzap is like a zap, except the zapper pays to *downweight* the zapped item in any search or sort rankings. If enough users downzap an item, it becomes “outlawed” and will be hidden from view for most users.<sup>5</sup>

Getting downzapped or outlawed is not a good outcome for the item’s creator. It lowers the item’s visibility and makes it less likely that the post will accumulate zaps. Since posting requires fees, this makes it more likely that the poster will lose money on the post. Moreover, getting downzapped lowers a user’s trust score, which reduces their ability to earn rewards or influence other users’ post rankings.

The requirement for downzaps to be paid with sats is an important feature of this system. It prevents users from frivolously downzapping other users and it prevents users from spinning up sybil accounts to do the same. By requiring a monetary payment for downvoting, Stacker News ensures that only content which is truly objectionable gets downzapped.

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<sup>5</sup>Users can opt in to seeing outlawed content in their settings.

**Hypotheses.** Based on the above discussion, we hypothesize the following about the incentive structures on Stacker News. First, we hypothesize that higher posting costs will lead to fewer posts in a territory, but higher quality posts. Second, we hypothesize that the types of content that are better rewarded by zaps will increase in prevalence over time, as users learn that those kinds of content are better rewarded. Third, we hypothesize that consistently unprofitable users are more likely to eventually leave the platform, whereas consistently profitable users are more likely to stay. We test these hypotheses one by one in the following sections of the paper.

### 3 Data

To test these hypotheses, we downloaded a snapshot of the Stacker News database on October 5, 2025. The data covers the time period from June 11, 2021 to October 5, 2025 and contains 207,219 posts and 914,266 comments (replies), for a total of 1,121,485 items posted by 11,228 users across 117 territories. Over this time period, users zapped each other 146 million sats and paid 13 million sats in posting fees.

In addition to the Stacker News data, we downloaded daily Bitcoin price data from CoinMarketCap.com. At the time of writing, the price of one Bitcoin is approximately US\$100,000, which would make 1,000 sats worth approximately \$1. This is a useful rule of thumb for converting between sats and dollars when trying to evaluate the economic significance of Stacker News micropayments.<sup>6</sup>

Figure 1 shows the growth in Stacker News, as measured by weekly number of items posted, alongside the weekly Bitcoin price. In aggregate, both the Bitcoin price and the activity on Stacker News grew over this time period.

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<sup>6</sup>In the subsequent text, whenever we convert sats to dollars, we will be using a conversion rate of 1,000 sats = \$1.

## 4 Pay-to-post and content quality

In this section, we consider our first hypothesis: whether higher posting costs lead to fewer posts in a territory, but higher quality posts. First, we develop a simple model of posting behavior to ground our empirical work. Then, we test whether the predictions of the model are supported by the data.

### 4.1 A model of pay-to-post

We envision a model in which a user receives “post ideas” as a Poisson arrival process with rate  $\gamma$ . Post ideas arrive with a quality level  $q$ , which affects how other users respond to the post, and an intrinsic motivation to post it,  $\epsilon$ . The level of internal motivation may be correlated with post quality, and they are drawn from a joint probability density function  $f(q, \epsilon)$ . We allow different users to draw from different distributions of  $(q, \epsilon)$ , but for now we focus on the decisions of a single user.

By Bayes’ Rule,  $f(q, \epsilon) = f_{\epsilon|q}(\epsilon|q)f_q(q)$ , where  $f_{\epsilon|q}(\cdot)$  is the conditional density of  $\epsilon$  given  $q$ , and  $f_q(\cdot)$  is the marginal density of  $q$ . We assume that  $f_{\epsilon|q}(\cdot)$  is strictly log-concave and weakly satisfies the monotone likelihood ratio property in  $\epsilon$  with respect to  $q$ :

**Assumption 1** (Log Concavity). *For each  $q$ ,  $f_{\epsilon|q}(\cdot|q)$  is strictly log-concave; that is,  $\ln f_{\epsilon|q}(\epsilon|q)$  is strictly concave in  $\epsilon$ .*

**Assumption 2** (Monotone Likelihood Ratio Property). *For any  $q_2 > q_1$ , the likelihood ratio of the conditional densities,*

$$\ell(\epsilon; q_1, q_2) = \frac{f_{\epsilon|q}(\epsilon|q_2)}{f_{\epsilon|q}(\epsilon|q_1)}$$

*is weakly increasing in  $\epsilon$ .*

*Remark.* Assumption 1 limits the heaviness of the tail of the motivation distribution. Intuitively, it ensures that the extreme tail of the motivation distribution is not thick enough to swamp quality selection effects at very high posting costs. It is not a strong assumption, and it is satisfied by most commonly used distributional families, including the normal and logistic distributions.<sup>7</sup>

Assumption 2 is a non-negative dependence condition between post quality  $q$  and intrinsic motivation  $\epsilon$ . Informally, it says that higher quality ideas tend to come with (weakly) higher intrinsic desire to post: when  $q$  is higher, the distribution of  $\epsilon$  shifts to the right. Independence between  $\epsilon$  and  $q$  is allowed as a special case, but the assumption rules out systematic negative dependence. In our context, this means that users who receive a high quality post idea are also (weakly) more likely to feel a strong intrinsic desire to post it. As we shall document later, this is consistent with users' self reported perceptions of their own motivations.  $\square$

Once a post is made, zaps arrive according to a Poisson process. Both the arrival rate of zaps and the amount of each zap are dependent on the quality of the post,  $q$ . Let  $\lambda(q)$  be the arrival rate as a function of  $q$ , and let  $E[z|q]$  be the expected zap amount per zap conditional on  $q$ .

Supposing the user has a discount factor of  $\rho$ , the expected present value of the Poisson stream of zaps is equal to:<sup>8</sup>

$$V(q) = \frac{\lambda(q)}{\rho} E[z|q] \tag{1}$$

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<sup>7</sup>Weaker, local conditions about the hazard rate in the neighborhood of the current posting cost would also suffice, but we state the stronger condition of global log-concavity for simplicity. Moreover, strict log-concavity is assumed for simplicity, but it suffices to assume either strict log-concavity or strict monotone likelihood ratio property; we do not need strictness for both.

<sup>8</sup>This follows from  $V(q) = \int_0^\infty e^{-\rho t} \lambda(q) E[z|q] dt$ .

We assume that  $V(q)$  is continuous and strictly increasing in  $q$ , which merely says that the expected present value of zaps is higher for higher quality posts:

**Assumption 3** (Monotone Returns to Quality).  $V(q) = \frac{\lambda(q)}{\rho} E[z|q]$  is continuous and strictly increasing in  $q$ .

The user will make the post if their expected present value of zaps plus their intrinsic motivation is greater than the posting cost  $c$ :

$$\text{Post} = 1 \text{ if } V(q) + \epsilon > c \text{ and } 0 \text{ otherwise} \quad (2)$$

From the perspective of the user,  $c$  is fixed and known at the time of the post.

We now demonstrate conditions under which an increase in posting cost  $c$  will lead to fewer posts, but higher quality posts.

**Proposition 1.** *Suppose Assumptions 1-3 hold. Then an increase in  $c$ :*

- (i) strictly decreases the probability that the post idea is posted, and*
- (ii) strictly increases the expected quality of posts, conditional on posting.*

*Proof.* A proof is provided in the appendix. □

*Remark.* Proposition 1 implies that when posting cost increases, users are less likely to post, but that when they do post, the average quality of the post is higher. Although the proposition is stated from the perspective of a single user, we note that as long as the global joint distribution of  $(q, \epsilon)$  holds to Assumptions 1 and 2, and if Assumption 3 also holds for all users, then Proposition 1 applies in the aggregate, even if users have heterogeneous marginal densities for  $q$  and  $\epsilon$ .

The result of Proposition 1 extends to a setting with multiple territories. Users post in the territory  $j$  that maximizes  $V_j(q_j) + \epsilon_j - c_j$ , provided it is positive. When territory  $j$  raises its posting cost  $c_j$ , users' outside options are unaffected, so the same logic applies: as long as the population-level joint distribution of quality and motivation within each territory satisfies Assumptions 1-3, a territory that raises its posting cost will see fewer but higher quality posts.  $\square$

## 4.2 Evidence on post quality

To empirically test the predictions of Proposition 1, we run regressions of the following form:

$$\ln \text{Quality}_{ijt} = \alpha \ln \text{Cost}_{ijt} + \delta_j + \gamma_t + \epsilon_{ijt} \quad (3)$$

where  $i$  indexes a post,  $j$  indexes a territory, and  $t$  indexes a year-week.<sup>9</sup>  $\text{Quality}_{ijt}$  is the measure of post quality,  $\text{Cost}_{ijt}$  is the cost (in sats) of the post,  $\delta_j$  is a set of dummy variables for each territory, and  $\gamma_t$  is a set of dummy variables for each week. The coefficient  $\alpha$  identifies the elasticity between posting cost and post quality.<sup>10</sup> We hypothesize that  $\alpha > 0$ .

To measure post quality, we measure the number of sats the post earned on zaps in the first 48 hours of posting. Zaps represent a revealed preference measure of quality, since users must pay real money to zap. The purpose of the dummy variables is to absorb any unobserved heterogeneity in territories (e.g. if some territories are more

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<sup>9</sup>For example, 2024w1 is a different  $t$  than 2025w1.

<sup>10</sup>In practice, we add 1 to post quality and posting cost to avoid taking logs of zero. Since the means of both variables are well above one, the  $\ln(1+x)$  transformation closely approximates  $\ln(x)$ , and the coefficients retain their interpretation as elasticities.

popular and thus attract more zaps), and to absorb any global time trends in the popularity of Stacker News.

The causal effect of posting cost is identified from differential changes in post quality across territories when the territory posting costs change differentially. For example, if the posting cost in the `econ` territory increases while the posting cost in `bitcoin` stays the same, and subsequently the average quality of posts increases in `econ` relative to that in `bitcoin`, this would suggest that the change in posting cost likely caused the change in post quality. The coefficient  $\alpha$  is identified off the many such changes present in the data.

To give the reader a sense of the variation in posting costs, we plot the posting cost histories for four representative territories in Figure 2. The figure shows that there is indeed substantial variation in posting fees both across and within territories.

Before running the regression, we apply the following sample selection criteria. We do not include posts in the `AMA` territory or the `jobs` territory because these territories seem to operate on different incentive structures than the other ones.<sup>11</sup> We also do not include a number of special post types that do not follow normal posting rules.<sup>12</sup> Lastly, we do not include any posts made in territories by the owner of the territory. Since 70% of posting costs go to the territory owner, the incentives for territory owners to post in their own territories differ from that of non-owners. In the data we use for the regression, the average posting cost is 51 sats, the average amount of sats earned in the first 48 hours is 364 sats, and the average number of comments in the first 48 hours is 3.3. There is substantial variation in all three: the

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<sup>11</sup>`AMA` is a territory where influential people can make “Ask Me Anything” posts. Because these posters are influential, posts in the `AMA` territory earn an unusually large amount of sats, and the posters are unlikely to be motivated by posting costs. `jobs` is a territory run by Stacker News in which companies can make job advertisements.

<sup>12</sup>These include profile posts, which users can post and edit for free, “saloon” posts, which are daily discussion threads posted by Stacker News, and freebies—a limited number of reduced-visibility posts that users are allowed to make for free.



standard deviation of the posting cost is 241 sats, of sats earned in the first 48 hours is 6,662, and of comments in the first 48 hours is 9.6.

Table 1 shows results for regressions of specification (3) where post quality is measured as zaps in the first 48 hours. Each column of Table 1 includes a different set of fixed effects in the regression. Column 1 includes no fixed effects, and is therefore a straight OLS regression of log zaps in the first 48 hours on log posting cost. Column 2 includes territory and week fixed effects to control for any heterogeneous effects between territories. Column 3 includes fixed effects for the identity of the territory owner to control for the zapping behavior of territory owners. For example, some territory owners zap every post that comes into their territory while others do not. Territory-owner fixed effects control for this behavior.<sup>13</sup> Finally, column 4 includes user fixed effects to test whether the gains in post quality are driven by within-user effects or compositional effects.

Table 1 shows that there is a statistically significant and robust positive relationship between territory posting fees and post quality as measured by zaps in the first 48 hours. In terms of the magnitude of the effect, Table 1 column 3 (our preferred specification) suggests that the elasticity between territory posting cost and expected post quality is 0.219. This roughly translates to: if a territory doubles its posting cost, then the quality of posts is expected to increase by 16.4 percent. The relationship is not explained by territory specific effects, such as a spurious correlation between posting fees and the popularity of a territory’s topic, nor is it explained by global time trends in posting fees and zap behavior on Stacker News. Nor is it explained by effects related to territory owners—for example, if territory owners who set high fees also tend to zap posts in their territory more generously. The best explanation for

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<sup>13</sup>Note that territory fixed effects do not fully absorb territory owner fixed effects, since a single user can own multiple territories, and territories can also change owners over time.

the correlation between posting cost and post quality is that users will only choose to post in a higher cost territory if they think that the post is of high enough quality to warrant the cost. This is consistent with Proposition 1 and with users’ self-reported perceptions of their own behavior on the platform. One Stacker News user is quoted saying:

“If I put a lot of effort into a post, I know that the posting fee will be offset by the zaps. And even if it doesn’t, I’m just happy to show the work I’ve done and I’m ready to pay for it. On the other hand, when I share a simple link with a few quick comments and quotes, it takes me a few minutes, and I look for the cheaper territory to post it. If I don’t find any cheap territory, I’d rather not post, as I don’t think paying to post for links is worth the same as paying to post for actual proof-of-work.”<sup>14</sup>

It makes sense that users will only be willing to pay a high posting cost if they think the quality of the post justifies it—either because they will recuperate the cost through zaps or because of a stronger intrinsic desire to post high quality work. However, one may wonder why users don’t just choose a low cost territory to post in. There is no rule on Stacker News that says you must post content to the most relevant territory. Moreover, most territories on Stacker News are lightly moderated, so posting content to an irrelevant territory is not likely to result in post removal. Despite these caveats, in practice we observe that users tend to post in territories for which the content is most relevant. This may be for a few reasons. First, users can subscribe to or mute territories.<sup>15</sup> Thus, posting in a more relevant territory will

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<sup>14</sup>In this statement, the user is referring to link posts, which are posts that include a link to an external URL, often without any text in the post body. Link posts usually receive less zaps and are considered to be of lower quality because the content is not produced by the user themselves, and are thus less rewarded by zaps.

<sup>15</sup>Subscribing to a territory means you will receive notifications when a new post is made to that territory. Muting a territory means that you will no longer see posts from that territory.

increase the visibility of your post to users who are most likely to be interested in it. Second, territory owners sometimes add value to their territories by collecting, summarizing, and rewarding top posts. Your post is less likely to be highlighted if it is not relevant to the territory. Lastly, users may have an intrinsic, aesthetic desire to post in the most relevant territory, especially if they view their content to be of high quality.<sup>16</sup>

Finally, we comment on column 4 of Table 1, where we include user fixed effects. The inclusion of user fixed effects reduces the estimated elasticity between posting cost and post quality from 0.219 to 0.091 (still statistically significant at the 5% level). This suggests that the increase in post quality in response to an increase in posting cost is driven by both intensive margin (within-user) and extensive margin (compositional) effects. On the intensive margin, the same users now post higher quality posts, conditional on posting. On the extensive margin, the composition of posters in the territory shifts towards higher quality, most likely due to lower quality users dropping out. Column 3 remains our preferred specification over column 4, because we view the effect on the composition of users as a dimension we want to capture when studying the relationship between posting cost and post quality.

#### 4.2.1 Event study analysis

Our empirical setting is similar to a difference-in-differences (DID) regression, because we are identifying the effect of territory fee changes from the change in post quality following the fee change, while allowing for baseline differences between territories and common time trends in post quality. In classical DID, an important assumption is that the average post quality in a territory does not systematically trend upwards prior to

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<sup>16</sup>In terms of the model, this would be reflected in a high positive affiliation between the territory specific components of  $(q_j, \epsilon_j)$ .

a post fee change. If it did so, that would call into question whether the timing of the fee change is endogenous to some prior effect that was already increasing quality. Our setting is a little more complicated than a classical DID because we have multiple fee changes in a territory, bidirectional fee changes, and continuous variation in the size of the fee change. Nevertheless, it would be useful for us to test whether there are any systematic post quality changes prior to a fee change.

To test this, we run the following regression:

$$\begin{aligned} \ln \text{Quality}_{ijt} = & \sum_{s=-12}^{12} \alpha_s^+ 1[\Delta \text{Cost}_{j,t-s} > 0] \\ & + \sum_{s=-12}^{12} \alpha_s^- 1[\Delta \text{Cost}_{j,t-s} < 0] + \delta_j + \gamma_t + \epsilon_{ijt} \end{aligned} \quad (4)$$

Here  $\Delta \text{Cost}_{jt}$  is a continuous variable for how much the posting cost in territory  $j$  changed in week  $t$ , and  $1[\Delta \text{Cost}_{jt} \geq 0]$  is an indicator for whether the posting cost in territory  $j$  increased/decreased in week  $t$ . Thus,  $\alpha_s^{+/-}$  measures the contribution to post quality of being  $s$  weeks away from a posting cost increase/decrease when  $s < 0$ , and the contribution to post quality of being  $s$  weeks after a posting cost increase/decrease when  $s > 0$ .

We estimate equation (4) and plot the estimates for  $\alpha_s^{+/-}$ , along with their 95% confidence intervals, in Figure 3. The left panels show the results for a specification in which user fixed effects were not included, and the right panels show the results for a specification in which user fixed effects were included.

Figure 3 shows that there was no systematic upward trend in post quality prior to post fee increases. After a post fee increase, we see that post quality begins to trend upwards. By comparing the top left and top right panels, we also see that including user fixed effects moderates the effect, but does not eliminate it—suggesting

the presence of both compositional and intensive margin effects. Although noisier, we find a similar result surrounding decreases in post fees. There is no systematic downward trend in post quality prior to post fee decreases, and following a post fee decrease post quality begins to trend slightly downwards.

### 4.3 Evidence on post quantity

Now we consider whether territory posting cost also influences the number of posts a territory receives. If users are sensitive to posting cost, then we expect that territory posting fee will be negatively correlated with number of posts. To test this, we run regressions of the following form:

$$\ln \text{Posts}_{jt} = \alpha \ln \text{Cost}_{jt} + \delta_j + \gamma_t + \epsilon_{jt} \quad (5)$$

where  $j$  indexes a territory and  $t$  indexes a week.  $\text{Posts}_{jt}$  is the number of posts posted in territory  $j$  in week  $t$ ,  $\text{Cost}_{jt}$  is territory  $j$ 's posting fee in week  $t$  (measured as the average across days),  $\delta_j$  is a set of territory fixed effects, and  $\gamma_t$  is a set of week fixed effects. The results of this regression, for various combinations of fixed effects, are presented in Table 2. As expected, there is a robust and statistically significant negative relationship between territory posting cost and the quantity of posts in a territory. Using column 3 as our preferred specification, the results imply an elasticity of post quantity with respect to post cost of  $-0.265$ . This translates to: if the posting cost doubles, the territory can expect to receive 16.8 percent fewer posts in a week.

### 4.3.1 Event study analysis

We now turn to an event study analysis to test whether there are systematic changes in post quantity prior to territory fee changes. We run the following regression:

$$\begin{aligned} \ln \text{Posts}_{jt} = & \sum_{s=-12}^{12} \alpha_s^+ 1[\Delta \text{Cost}_{j,t-s} > 0] \\ & + \sum_{s=-12}^{12} \alpha_s^- 1[\Delta \text{Cost}_{j,t-s} < 0] + \delta_j + \gamma_t + \epsilon_{ijt} \end{aligned} \quad (6)$$

The notation and logic is much the same as in (4), so we will not repeat ourselves here.

Figure 4 shows the estimated values for  $\alpha_s^{+/-}$  and their 95% confidence intervals.

Figure 4 shows that there is no systematic decrease in post quantity prior to a territory fee increase. If anything, there appears to be a systematic upward trend in post quantity prior to fee increases, followed by a downward trend in post quantity after the fee increase. For negative fee changes, there appears to be a slight systematic downward trend in post quantity prior to a fee decrease, followed by an upward trend in post quantity after a fee decrease.

Comparing Figures 3 and 4 reveals a pattern that is consistent with our hypothesized mechanism. Prior to a fee increase, post quantity trends upwards while post quality remains flat. The absence of a pre-trend in quality indicates that quality improvements observed after a fee increase are not the continuation of a pre-trend that was already underway. Rather, quality begins to rise only after the fee change takes effect, consistent with the screening mechanism described in Proposition 1. The rising quantity prior to a fee increase suggests that territory operators may be raising fees in response to growing volume, with the quality improvement emerging as a consequence of the higher fees screening out lower quality posts. The reverse pattern surrounding fee decreases tells a symmetric story. This story appears consistent with

how some territory operators on the platform have described their rationale for fee changes. One territory operator is quoted saying:

“For **science**, I only experimented a very little at the beginning. Don’t remember the specifics, but I must have tried really low (like 10-20 sats) and then a bit higher (like 100 sats). Because I had a personal policy of upvoting each post that had a modicum of quality (e.g. not just a link without context), I ended up with a small army of posters (assmilkers) that were doing the bare minimum but not enough to instigate comments or discussions. I think the reason for the 100 sats was to see how this increase would affect the willingness of those low-effort posters to post. It worked, but a side effect is that I was getting much less volume, also from quality posters. So I put it back down, and just let the community decide if a post is worth getting sats.”

This operator’s reasoning reflects the pay-to-post mechanism we described in Proposition 1.

## 4.4 Discussion

Taken together, the results are both surprising and unsurprising. They are unsurprising in the sense that they conform with economic theory: demand curves slope downwards (when posting costs go up, number of posts goes down), and signaling theory works (when posting costs go up, higher quality posts are made). They are surprising in the sense that even such small micro-incentives (the average posting cost is just 51 sats, or about 5 cents) are enough to influence user behavior in such a way that post quality is improved. The results suggest that even very small posting fees

can meaningfully affect the quantity-quality tradeoff in user-generated content. Moreover, the assumptions required for this result to hold more broadly are fairly weak: we require only that expected rewards are increasing in post quality (Assumption 3), that post quality and intrinsic motivation to post are positively affiliated (Assumption 2), and that extreme levels of intrinsic motivation are sufficiently rare (Assumption 1). All three assumptions are likely to hold on other platforms where micropayment mechanisms could be introduced, suggesting that these findings may generalize beyond our specific setting. However, it should be acknowledged that Stacker News is a highly selected group of individuals organized around a shared interest in Bitcoin, so caution is advised if extrapolating the magnitude of our results to other settings.

## 5 Value-for-value incentives and content quality

We now consider our second hypothesis: that the types of content better rewarded by zaps will increase in prevalence over time as users learn that these kinds of content are better rewarded. As in the previous section, we first extend the model of posting behavior to account for effort and learning, then we test the empirical predictions of the model.

### 5.1 A model with learning and effort

We extend the model from Section 4 to account for effort. As before, opportunities to post still arrive as a Poisson process with rate  $\gamma$ . But before observing  $(q, \epsilon)$ , the user first decides how much effort to exert in crafting the post. Effort affects the distribution of post quality and intrinsic motivation,  $f(q, \epsilon; e)$  (users may be more motivated to post if they’ve exerted a lot of effort). The cost of exerting effort is  $\kappa(e)$



with  $\kappa(0) = 0$  and  $\kappa'(e) > 0$ . The user's expected utility conditional on effort is:

$$U(e) = E_{q,\epsilon|e} [\max \{V(q) - c + \epsilon, 0\}] - \kappa(e)$$

Without uncertainty, the model would be equivalent to the one presented in Section 4, just with the distribution of  $(q, \epsilon)$  moderated by the optimal effort level. We therefore assume that users have uncertainty over the returns to quality. That is, let:

$$V(q) = \exp(\theta + q) \tag{7}$$

where users have uncertainty over  $\theta$ . Note that this is without loss of generality because  $q$  has no natural units.

Given that users are uncertain about  $\theta$ , they must infer it from their received zaps relative to quality. We assume that users observe  $q$ . Let  $z_i$  be the zaps received within a fixed time period, e.g. 48 hours, for a specific post  $i$ .  $z_i$  is a noisy signal proportional to  $V(q_i)$ , and therefore:

$$\ln z_i = \theta + q_i + \text{noise} \tag{8}$$

Now let a user have a posting history of posts indexed by  $i = 1, \dots, N$ . An unbiased estimator for  $\theta$  is:

$$\hat{\theta} = \frac{1}{N} \sum_{i=1}^N \ln z_i - q_i \tag{9}$$

That is, the user's unbiased estimate for  $\theta$  is the average of their posts' zap "surprise", i.e. how much additional zapping it received over and above its baseline quality level.<sup>17</sup>

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<sup>17</sup>This assumes diffuse priors; if the user had a more precise prior, their posterior mean would still be linear in  $\frac{1}{N} \sum \ln z_i - q_i$ .

We can now demonstrate that effort exertion is increasing in  $\hat{\theta}$  and that, as a corollary, the expected quality of the next post is increasing in  $\hat{\theta}$ .

**Proposition 2.** *Let  $i = 1, \dots, N$  index a user's previously made posts. Let  $q_i$  denote the quality of those posts, let  $z_i \propto V(q_i)$  be a noisy signal proportional to the present value of zaps that post  $i$  receives, and let  $\hat{\theta} = \frac{1}{N} \sum \ln z_i - q_i$ . Furthermore, assume that  $F(q|\epsilon; e)$  satisfies FOSD in  $e$ , conditional on  $\epsilon$ . Then, optimal effort is weakly increasing in  $\hat{\theta}$ .*

*Proof.* A proof is provided in the appendix. □

## 5.2 Evidence of learning

Since  $\hat{\theta}$  is the average of the zap surprises across previous posts, Proposition 2 suggests that we can test for the presence of learning by regressing the quality of a user's next post on the average zap surprise for all prior posts made by that user. Specifically, let  $i$  index a user and let  $k$  index the user's  $k$ th post. We would run regressions of the following form:

$$q_{i,k+1} = \beta_0 + \beta_1 \underbrace{\frac{1}{k} \sum_{j=1}^k [\ln z_{ij} - q_{ij}]}_{\hat{\theta}_{ik}} + \epsilon_{i,k+1} \quad (10)$$

where  $z_{ij}$  is the zaps user  $i$  received on their  $j$ th post in the first 48 hours, and  $q_{ij}$  is the quality of user  $i$ 's  $j$ th post. If users are learning about the returns to effort from past zap surprises, we would expect to estimate  $\beta_1 > 0$ . Identification comes from the following thought experiment: imagine two otherwise identical users, but by random chance user A has had a more positive zap history than user B for the same quality posts. User A therefore has a more optimistic view of the returns to effort

than user B, and will on the subsequent post exert more effort, resulting in higher expected post quality.

To measure  $q_{ij}$ , we make use of equation (8), which implies that the zap amount within any fixed time window is a noisy measure of  $q_{ij}$ . We can therefore estimate  $q_{ij}$  for each post by regressing zaps on objectively measured post attributes, such as number of words, whether the post is a link post, the number of images and links in the content, etc:

$$\ln z_{ij} = \theta + \underbrace{X_{ij}\beta}_{q_{ij}} + \varepsilon_{ij} \quad (11)$$

The model’s predicted values of  $X_{ij}\hat{\beta}$  can then be used as unbiased estimates of the post quality  $q_{ij}$ .

Table 3 reports results from regressions based on equation (11). Column 1 starts with three attributes: the number of words in the post, whether or not the post is a link post, and the number of images and links in the post body. As discussed earlier, link posts are posts that point to a URL. They can have empty post bodies (no words), but the author of a link post can also add text to the post body to provide context for the link. As confirmed by the results in Table 3, link posts are generally less well-rewarded on Stacker News than non-link posts. By contrast, posts with links and images *in the post body* are usually better rewarded on average than posts without links and images. The presence of images and links in the post body may signal that the author has spent some time thinking about and working on the post, both to make it more attractive and engaging (images), and to make it more useful (referrals to other relevant URLs).

Column 2 of Table 3 adds text embeddings to the vector of post attributes. The embeddings were computed from post text using OpenAI’s `text-embedding-3-small`

embedding model. The embedding model computes a 1,536 dimensional embedding vector for each post that encodes the meaning of the text in such a way that two posts with similar text will have short distances between their embedding vectors. Before including these embedding vectors in the vector of attributes, we first reduce their dimensionality by performing principal components analysis (PCA) with 10 components. The PCA extracts the 10 orthogonal component vectors that explain the most variance in the original 1,536 dimensional vector. We then include the 10 principal components in the attribute vector  $X_{ij}$  for the regression.<sup>18</sup>

Lastly, Column 3 of Table 3 includes user fixed effects and territory fixed effects. It is appropriate to include these features in our definition of  $q_{ij}$ , because  $q_{ij}$  is meant to capture all the predictable features of post quality, and it is reasonable to assume that users have a sense of which territories are better rewarded and a sense of the baseline quality of their own writing.

Column 3 of Table 3 is our preferred specification. We therefore take the coefficients as estimated in Column 3 of Table 3 and estimate  $q_{ij}$  for each post using only those attributes. We then regress  $q_{i,k+1}$  on  $\hat{\theta}_{ik}$ , i.e. we regress the objective quality metric of the next post on the average of past posts' zap residuals, as suggested in equation (10). Table 4 shows the results of this regression. Note that user and territory fixed effects are included in the regression by default because we are trying to control for two otherwise identical users who have different histories of zap surprises. In Column 2 of Table 4, we additionally include the user's history of total prior zaps and total prior posts, to further control for time-varying user heterogeneity. Lastly, in Column 3 we include week fixed effects to control for global time trends in the quality of posts. Global time trends could itself be due to learning: especially if users learn not just from their own posts but from the posts of other users. Controlling

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<sup>18</sup>Please refer to the appendix for further methodological details.

for global time trends therefore isolates the effect of  $\beta_1$  to the individualized learning that a user experiences from their personal zap history.

Using Column 3 of Table 4 as our preferred specification, the results show an elasticity of 0.023 between subsequent post quality and the average of past zap surprises. The effect is modest: it implies that if user A has twice the average past zap surprise of user B, then user A’s subsequent post will be 1.6 percent higher quality, measured by expected zaps received. Although the magnitude of the effect is small, the effect is statistically significant and robust to the inclusion of various controls, as shown in Table 4.

### 5.3 Discussion

Although the magnitude of our estimated learning effect is small, we note that this regression estimates learning driven by variations in individual experiences only, and partials out the effect of global learning via the week fixed effects. If we hypothesize that global learning is also present—that is, if users are learning from which posts made by *other* users are highly rewarded (which is almost certainly true)—then we should expect to see higher post quality over time globally, as measured by objective post attributes. And this is indeed what we see: Figure 5 plots the average of  $q_{ij}$  by week, showing that this has indeed been increasing over time. In other words, post quality—as measured by number of words and semantic features—has been increasing over time, which is suggestive of global learning and quality improvement driven by value-for-value incentives. Note that this result is *not* driven by changes to zap generosity over time, since  $q_{ij}$  is measured based on objective post attributes only. We do acknowledge, however, that any effect of global learning is hard to disentangle from alternative explanations that could drive a similar effect, such as a global change

in posters’ intrinsic preferences (e.g. if posters’ intrinsic desire to make longer posts has grown over time.)

## 6 Profitability and user retention

We now test our last hypothesis: that consistently unprofitable users are more likely to exit the platform, whereas profitable users are more likely to stay. This would arise straightforwardly out of the learning model in which some users learn that their returns to effort are consistently low.

To measure profitability, we only consider whether the user is able to earn enough zaps and rewards to cover their posting costs. We do not consider their own zaps to other users as part of their costs, since zapping is entirely optional. Moreover, some users may have altruistic motives to zap, as discussed in the introduction, so greater zapping could indicate a greater commitment to the platform.

We define a user as inactive if they go four or more consecutive weeks without making any posts, any comments, or zapping any items. The first week of inactivity is considered the week they went inactive. Inactive users can become active again, though we treat this event as exogenous.

The goal of this section is to estimate how user profitability affects the probability of becoming inactive. We estimate linear probability models of the following form:

$$\text{Inactive}_{it} = \alpha \text{Unprofitable}_{it} + \beta \text{Unprofitable}_{it} \times \text{Growth}_t + \gamma_t + \epsilon_{it} \quad (12)$$

where  $i$  indexes a user and  $t$  indexes a week.  $\text{Inactive}_{it}$  is a binary indicator for whether user  $i$  became inactive on week  $t$ ,  $\text{Unprofitable}_{it}$  is a binary indicator for whether the user was unprofitable in the last 8 weeks, and  $\text{Growth}_t$  is the percentage

growth in Bitcoin-USD price in the last 8 weeks.  $\gamma_t$  is a week fixed effect to capture global time trends, and  $\epsilon_{it}$  is the error term. We hypothesize that  $\alpha > 0$ , indicating that users who have been unprofitable are more likely to become inactive, and that  $\beta > 0$ , that this effect is amplified when Bitcoin’s price has recently appreciated.

Table 5 reports the results for this regression. Column 1 shows the result when only profitability in the last 8 weeks is included as a regressor. Column 2 additionally controls for the number of items the user posted in the last 8 weeks. Unsurprisingly, users who have posted more items in the last 8 weeks are less likely to become inactive, as it shows a greater commitment to the platform. Including the number of items in the last 8 weeks significantly moderates the effect of unprofitability, suggesting that there are some unprofitable users who post a lot anyway. Column 3 adds week fixed effects to control for global time trends, but the coefficient of interest is not much changed. Lastly, column 4 interacts unprofitability with the Bitcoin price appreciation over the last 8 weeks.

Using column 4 as the preferred specification, the results show that users who were unprofitable in the last 8 weeks are 5.5 percentage points more likely to become inactive than users who were profitable in the last 8 weeks. This is in comparison to a baseline inactive probability of 14.1% per week. In addition, column 4 shows that unprofitable users are even more likely to become inactive when the Bitcoin price has recently appreciated. If the Bitcoin price has appreciated by 10 percent in the last 8 weeks, this increases the effect of unprofitability from a 5.5 percentage-point to a 6.1 percentage-point increased probability of exit. This suggests that users are sensitive to not just their profitability as measured in sats, but also their profitability as measured in purchasing power.

## 7 Conclusion

Our analysis of Stacker News data confirms three central hypotheses: first, that posting costs can increase the quality of posted content, consistent with signaling theory; second, that the potential to be rewarded with tips also leads users to post better content over time; and third, that unprofitable users are more likely to leave the platform while profitable users are more likely to stay. This third point suggests that content quality may further increase over time as users who consistently produce low-value content eventually exit the platform.

More broadly, the results suggest that users of digital platforms are responsive to financial micro-incentives, even ones of very small monetary value. Integrating micro-payment systems into user-generated content platforms could therefore be an effective approach to managing the quantity-quality tradeoff in content production. Our estimated elasticities provide a starting point for platform designers seeking to calibrate such mechanisms. As low-cost payment infrastructure like the Lightning Network become more widely available, we expect these design tools to become increasingly practical for a broad range of digital platforms.

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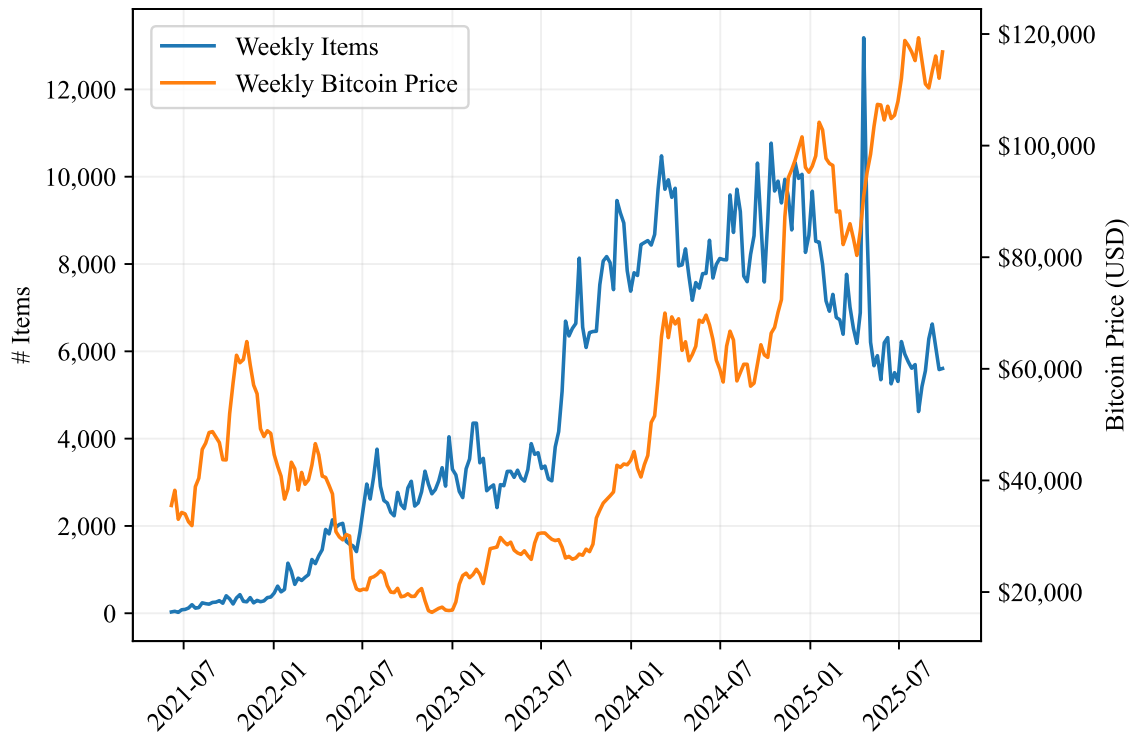
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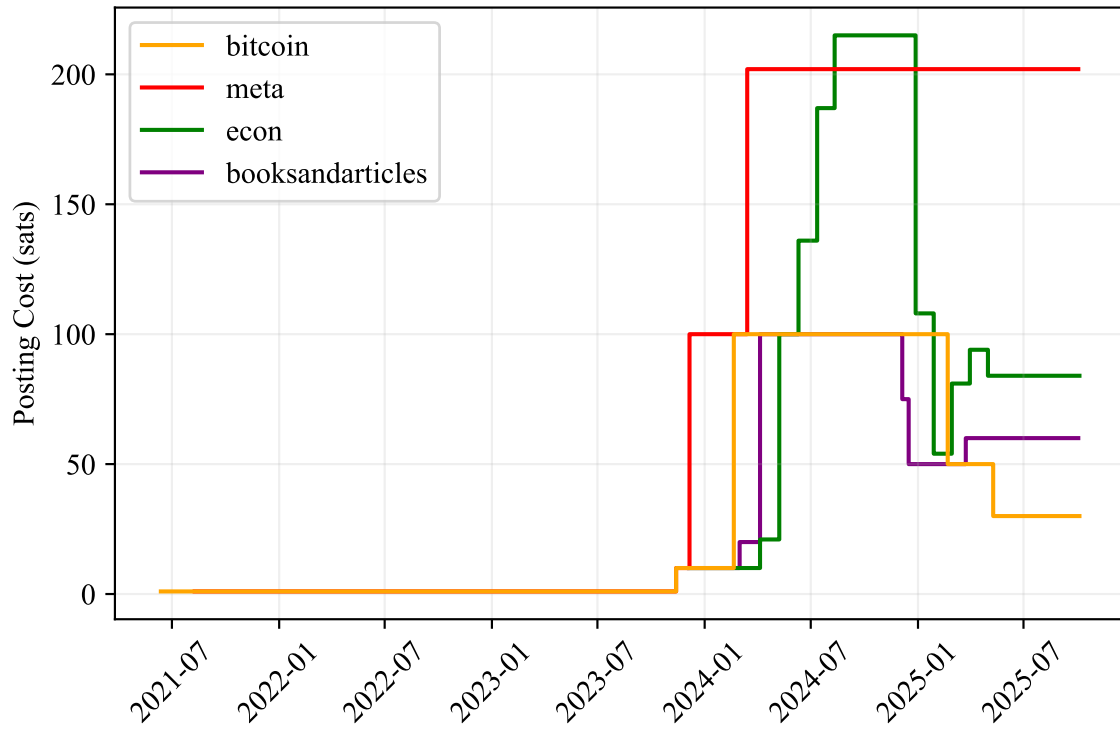
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Figure 1: Weekly Stacker News Items and Bitcoin Price



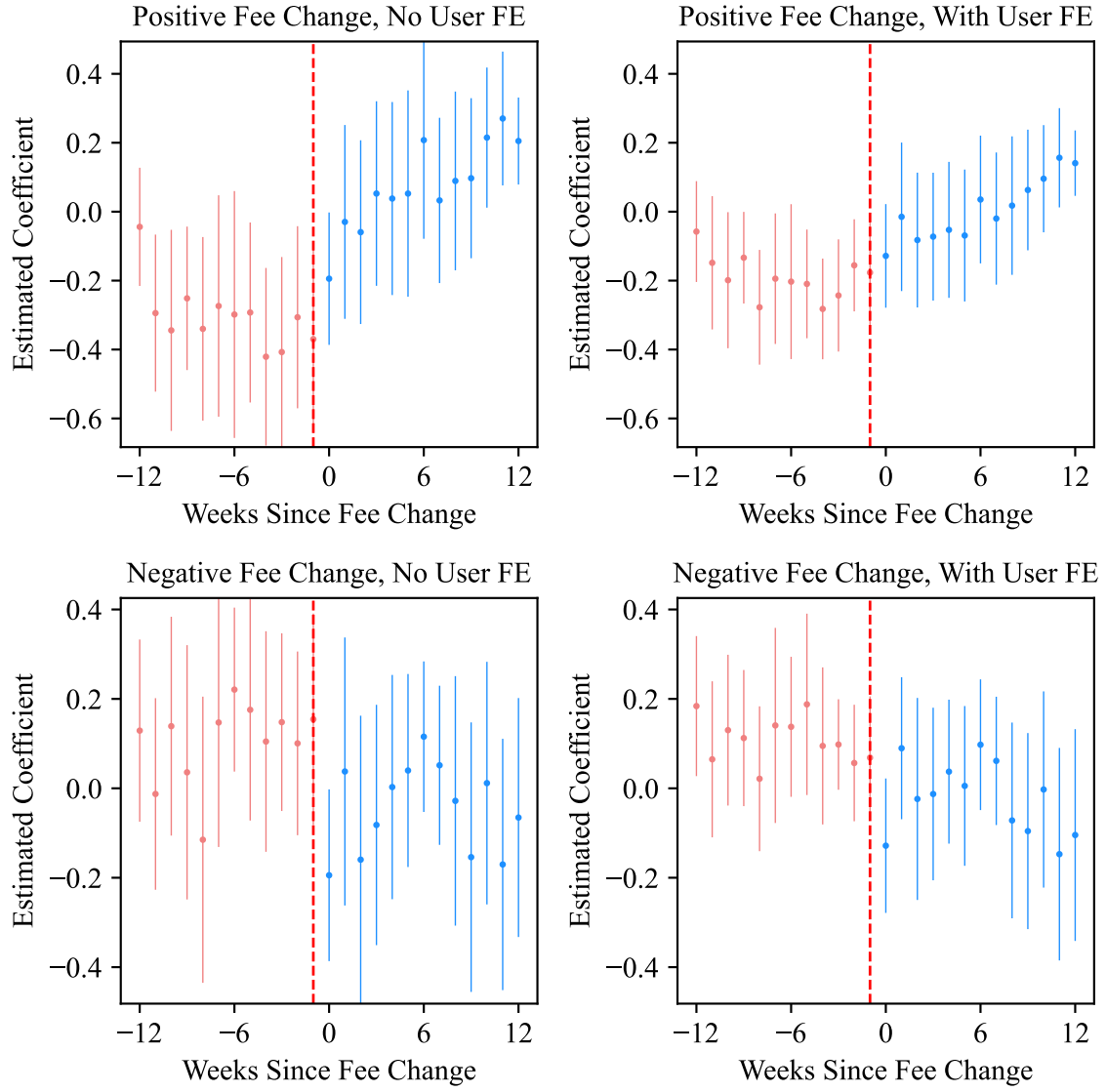
*Note:* Shows the number of items posted on Stacker News by week along with the weekly Bitcoin price. Weekly Bitcoin price is calculated as the average of the midpoint between daily highs and lows as reported by CoinMarketCap.com.

Figure 2: Posting Cost Histories for Select Territories



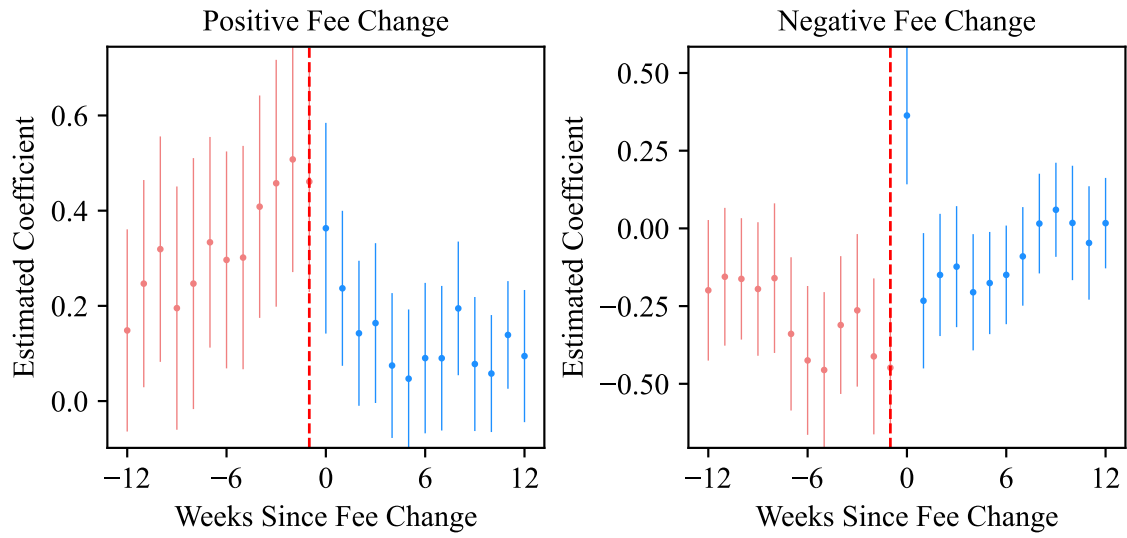
*Note:* This figure shows the territory fee path for four selected territories. The territories were selected both for their popularity and to highlight the differences in timing and magnitude of fee changes across territories.

Figure 3: Post Quality Event Study



*Note:* Shows estimated coefficients and 95% confidence intervals from the post quality event study regression specified by equation (4). All regressions shown include territory-owner fixed effects and week fixed effects.

Figure 4: Post Quantity Event Study



*Note:* Shows estimated coefficients and 95% confidence intervals from the post quantity event study regression specified by equation (6). All regressions shown include territory fixed effects and week fixed effects.

Figure 5: Post Quality Over Time



*Note:* Shows the average post quality as measured by objective post attributes ( $q_{ij}$ ) for each week. See Section 5 for details.

Table 1: Effect of Posting Cost on Post Quality

|                       | <i>Dependent variable:</i> |                     |                     |                    |
|-----------------------|----------------------------|---------------------|---------------------|--------------------|
|                       | log(Zaps)                  |                     |                     |                    |
|                       | (1)                        | (2)                 | (3)                 | (4)                |
| log(Posting Cost)     | 0.393***<br>(0.048)        | 0.273***<br>(0.058) | 0.219***<br>(0.050) | 0.091**<br>(0.044) |
| Constant              | 2.530***<br>(0.076)        |                     |                     |                    |
| Territory FE          | N                          | Y                   | Y                   | Y                  |
| Week FE               | N                          | Y                   | Y                   | Y                  |
| Territory Operator FE | N                          | N                   | Y                   | Y                  |
| User FE               | N                          | N                   | N                   | Y                  |
| Observations          | 179,480                    | 179,477             | 179,476             | 178,197            |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

*Notes:* Regression of number of sats a post earns in the first 48 hours on posting cost. The unit of observation is a post. Standard errors are clustered by territory.



Table 2: Effect of Posting Cost on Post Quantity

|                       | <i>Dependent variable:</i> |                      |                      |
|-----------------------|----------------------------|----------------------|----------------------|
|                       | log(Posts)                 |                      |                      |
|                       | (1)                        | (2)                  | (3)                  |
| log(Posting Cost)     | -0.242***<br>(0.079)       | -0.266***<br>(0.063) | -0.265***<br>(0.074) |
| Constant              | 3.328***<br>(0.415)        |                      |                      |
| Territory FE          | N                          | Y                    | Y                    |
| Week FE               | N                          | Y                    | Y                    |
| Territory Operator FE | N                          | N                    | Y                    |
| Observations          | 6,023                      | 6,020                | 6,020                |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

*Notes:* Regression of number of posts in a territory on posting costs for that week. The unit of observation is a territory-week. Standard errors are clustered by territory.

Table 3: Regression of Post Zaps on Post Attributes

|                                 | <i>Dependent variable:</i> |                     |                      |
|---------------------------------|----------------------------|---------------------|----------------------|
|                                 | log(Zaps)                  |                     |                      |
|                                 | (1)                        | (2)                 | (3)                  |
| log(# Words)                    | 0.354***<br>(0.054)        | 0.266***<br>(0.085) | 0.142***<br>(0.038)  |
| No text                         | 0.193<br>(0.255)           | -0.378<br>(0.295)   | -0.139*<br>(0.082)   |
| Is link post                    | -0.531**<br>(0.239)        | -0.211**<br>(0.098) | -0.410***<br>(0.059) |
| # Images and links in post body | 0.019*<br>(0.011)          | 0.014**<br>(0.006)  | 0.009*<br>(0.005)    |
| Constant                        | 3.124***<br>(0.403)        | 3.380***<br>(0.461) |                      |
| Text Embeddings                 | N                          | Y                   | Y                    |
| Territory FE                    | N                          | N                   | Y                    |
| User FE                         | N                          | N                   | Y                    |
| Observations                    | 175,857                    | 175,857             | 174,578              |
| $R^2$                           | 0.155                      | 0.181               | 0.404                |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

*Notes:* Regression of number of zaps a post earns in the first 48 hours on post attributes. The unit of observation is a post. Standard errors are clustered by territory.

Table 4: Regression of Next Post Quality on Past Zap Surprises

|                                                   | <i>Dependent variable:</i>           |                     |                      |
|---------------------------------------------------|--------------------------------------|---------------------|----------------------|
|                                                   | Quality of next post ( $q_{i,k+1}$ ) |                     |                      |
|                                                   | (1)                                  | (2)                 | (3)                  |
| Avg of past zap surprises ( $\hat{\theta}_{ik}$ ) | 0.048***<br>(0.007)                  | 0.033***<br>(0.009) | 0.023***<br>(0.007)  |
| log(Total prior zaps)                             |                                      | 0.017***<br>(0.004) | -0.012***<br>(0.003) |
| log(Total prior posts)                            |                                      | 0.000***<br>(0.000) | -0.000<br>(0.000)    |
| Territory FE                                      | Y                                    | Y                   | Y                    |
| User FE                                           | Y                                    | Y                   | Y                    |
| Week FE                                           | N                                    | N                   | Y                    |
| Observations                                      | 170,936                              | 170,936             | 170,936              |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

*Notes:* Regression of next post quality ( $q_{i,k+1}$ ) on the average of prior zap surprises ( $\hat{\theta}_{ik} = \frac{1}{k} \sum_{j=1}^k \ln z_j - q_j$ ). The unit of observation is a post. Standard errors are clustered by territory.

Table 5: Effect of Profitability on User Exit

|                                                       | <i>Dependent variable:</i>    |                      |                      |                      |
|-------------------------------------------------------|-------------------------------|----------------------|----------------------|----------------------|
|                                                       | Whether user becomes inactive |                      |                      |                      |
|                                                       | (1)                           | (2)                  | (3)                  | (4)                  |
| Unprofitable in last 8 weeks                          | 0.105***<br>(0.003)           | 0.062***<br>(0.003)  | 0.060***<br>(0.003)  | 0.055***<br>(0.003)  |
| $\dots \times$ BTC price appreciation in last 8 weeks |                               |                      |                      | 0.056***<br>(0.014)  |
| $\log(\text{Items posted in last 8 weeks})$           |                               | -0.029***<br>(0.001) | -0.029***<br>(0.001) | -0.029***<br>(0.001) |
| Constant                                              | 0.079***<br>(0.001)           | 0.157***<br>(0.002)  |                      |                      |
| Week FE                                               | N                             | N                    | Y                    | Y                    |
| Observations                                          | 92,956                        | 92,956               | 92,956               | 92,956               |

\*  $p < 0.1$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

*Notes:* Linear probability model of whether a user becomes inactive in a given week. A user becomes inactive if they start a four or more week consecutive period of no activity. The unit of observation is a user-week.

# Online Appendix

## A Proofs

### Proof of Proposition 1

*Proof.* A post idea  $(q, \epsilon)$  is posted if and only if  $\epsilon > c - V(q)$ . Define  $\pi(q, c)$  as the probability (over  $\epsilon$ ) that a post of quality is posted when the cost is  $c$ :

$$\pi(q, c) = P(\epsilon > c - V(q))$$

$\pi(q, c)$  is equivalent to the survival function of  $\epsilon|q$ , evaluated at  $c - V(q)$ :  $\pi(q, c) = S_{\epsilon|q}(c - V(q)|q)$ .

(i) The unconditional probability of posting is:

$$P(\text{post}) = \int S_{\epsilon|q}(c - V(q)|q) f_q(q) dq$$

Since  $S_{\epsilon|q}(\cdot|q)$  is a survival function, its first derivative is negative everywhere. An increase in  $c$  will therefore reduce the value of  $S_{\epsilon|q}(c - V(q))$  at every  $q$ . Since that is what we are integrating over to get  $P(\text{post})$ , this strictly reduces  $P(\text{post})$ . The expected number of posts per unit time is  $\gamma P(\text{post})$ , so an increase in  $c$  strictly reduces the expected number of posts.

(ii) Take any  $c' > c$ . The set of ideas that were posted at cost  $c$  can be partitioned into two groups:

- **Surviving posts.** Ideas that are still posted at cost  $c'$ , i.e.  $V(q) + \epsilon > c'$
- **Marginal posts.** Ideas that are posted at  $c$  but not at  $c'$ , i.e.  $c < V(q) + \epsilon \leq c'$

$$V(q) + \epsilon < c'.$$

By the Law of Iterated Expectations,

$$E[q|\text{post at } c] = \alpha E[q|\text{surviving}] + (1 - \alpha)E[q|\text{marginal}]$$

where  $\alpha = P(\text{surviving}|\text{post at } c)$ . Noting that  $E[q|\text{post at } c'] = E[q|\text{surviving}]$ , we rearrange the above equation to obtain:

$$E[q|\text{post at } c'] - E[q|\text{post at } c] = (1 - \alpha) (E[q|\text{surviving}] - E[q|\text{marginal}])$$

Thus, to show that expected post quality is increasing in  $c$ , it suffices to show that the expected quality of the surviving posts is always higher than the expected quality of the marginal posts.

To demonstrate this, it is sufficient to establish that for any  $q_2 > q_1$ , the ratio  $\pi(q_2, c)/\pi(q_1, c)$  is strictly increasing in  $c$ . Intuitively, this says that as costs rise, higher quality posts retain a larger share of the posting probability relative to lower quality ideas. This would imply that the marginal posts are tilted towards lower quality than surviving posts.

We write:

$$\frac{\pi(q_2, c)}{\pi(q_1, c)} = \frac{S_{\epsilon|q}(c - V(q_2)|q_2)}{S_{\epsilon|q}(c - V(q_1)|q_1)}$$

Define the hazard rate:  $h(\epsilon|q) = f_{\epsilon|q}(\epsilon|q)/S_{\epsilon|q}(\epsilon|q)$ . Differentiating  $\pi(q_2, c)/\pi(q_1, c)$  with respect to  $c$  shows us that the ratio is increasing in  $c$  if and only if:

$$h(c - V(q_1)|q_1) > h(c - V(q_2)|q_2)$$

The proof now proceeds in two parts.

**Claim 1.** First, we show that  $h(\epsilon|q_1) \geq h(\epsilon|q_2)$  for any  $q_2 > q_1$ . By the MLRP, for any  $t > \epsilon$ , we have that  $\ell(t; q_1, q_2) \geq \ell(\epsilon; q_1, q_2)$ , which rearranges to:

$$f_{\epsilon|q}(t|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq f_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(t|q_1)$$

Integrating both sides over  $t$  from  $\epsilon$  to  $\infty$  gives us:

$$S_{\epsilon|q}(\epsilon|q_2)f_{\epsilon|q}(\epsilon|q_1) \geq S_{\epsilon|q}(\epsilon|q_1)f_{\epsilon|q}(\epsilon|q_2)$$

which rearranges to  $h(\epsilon|q_1) \geq h(\epsilon|q_2)$  as was to be demonstrated.

**Claim 2.** Now, we show that  $h(\epsilon|q)$  is increasing in  $\epsilon$  for each  $q$ . Take the derivative of  $h(\epsilon|q)$  with respect to  $\epsilon$ . The sign is equal to the sign of:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q)S_{\epsilon|q}(\epsilon|q)$$

When  $f'_{\epsilon|q}(\epsilon|q) \geq 0$ , this is immediately positive. When  $f'_{\epsilon|q}(\epsilon|q) < 0$ , we rely on the log-concavity property. Since  $f_{\epsilon|q}(\cdot)$  is log-concave, the following holds for every  $t > \epsilon$ :

$$\ln f_{\epsilon|q}(t|q) < \ln f_{\epsilon|q}(\epsilon|q) + (t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)}$$

Exponentiating both sides:

$$f_{\epsilon|q}(t|q) < f_{\epsilon|q}(\epsilon|q) \exp \left[ (t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right]$$

Now integrating over  $t$  from  $\epsilon$  to  $\infty$ :

$$\begin{aligned} S_{\epsilon|q}(\epsilon|q) &< f_{\epsilon|q}(\epsilon|q) \int_{\epsilon}^{\infty} \exp \left[ (t - \epsilon) \frac{f'_{\epsilon|q}(\epsilon|q)}{f_{\epsilon|q}(\epsilon|q)} \right] dt \\ &= - \frac{f_{\epsilon|q}(\epsilon|q)^2}{f'_{\epsilon|q}(\epsilon|q)} \end{aligned}$$

Since  $f'_{\epsilon|q}(\epsilon|q) < 0$ , this can be rearranged to:

$$f_{\epsilon|q}(\epsilon|q)^2 + f'_{\epsilon|q}(\epsilon|q) S_{\epsilon|q}(\epsilon|q) > 0$$

which was to be demonstrated. Thus,  $h(\epsilon|q)$  is strictly increasing in  $\epsilon$  for all  $q$ .

By the monotonicity assumption,  $c - V(q_1) > c - V(q_2)$ . By Claim 2, this implies that  $h(c - V(q_1)|q_1) > h(c - V(q_2)|q_1)$ . By Claim 1,  $h(c - V(q_2)|q_1) \geq h(c - V(q_2)|q_2)$ . Together, we have  $h(c - V(q_1)|q_1) > h(c - V(q_2)|q_2)$ , which was to be demonstrated.

□

## Proof of Proposition 2

*Proof.* Given  $\hat{\theta}$ , the user's expected utility from effort is:

$$U(e; \hat{\theta}) = E_{q, \epsilon|e} \left[ \max \left\{ \exp(\hat{\theta} + q) - c + \epsilon, 0 \right\} \right] - \kappa(e)$$

It suffices to show that  $U(e; \hat{\theta})$  is supermodular in  $(e, \hat{\theta})$ . For any  $\hat{\theta}_2 > \hat{\theta}_1$ :

$$\begin{aligned} U(e; \hat{\theta}_2) - U(e; \hat{\theta}_1) &= \\ &E_{q, \epsilon|e} \left[ \max \left\{ \exp(\hat{\theta}_2 + q) - c + \epsilon, 0 \right\} - \max \left\{ \exp(\hat{\theta}_1 + q) - c + \epsilon, 0 \right\} \right] \end{aligned}$$



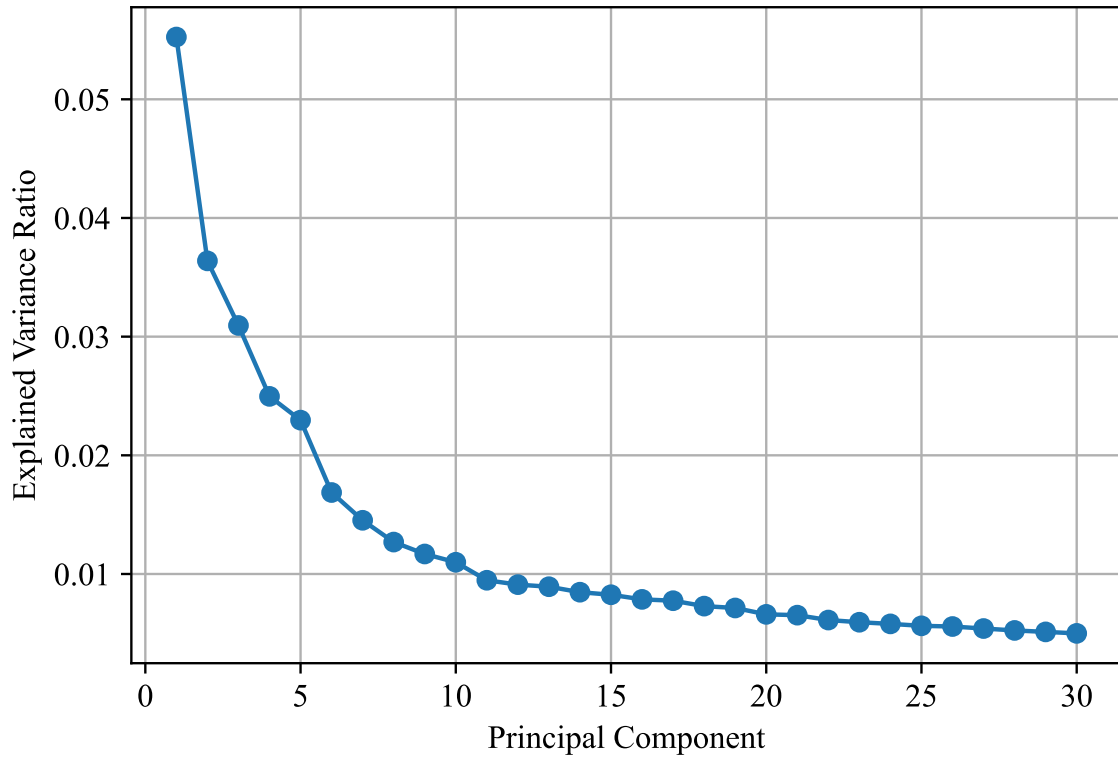
The integrand is supermodular in  $(\hat{\theta}, q)$  because  $\exp(\hat{\theta} + q)$  is supermodular in  $(\hat{\theta}, q)$  and because the max operator is convex and increasing. Furthermore, since  $F(q|\epsilon; e)$  satisfies FOSD in  $e$ , it follows immediately  $U(e; \hat{\theta}_2) - U(e; \hat{\theta}_1)$  is increasing in  $e$ , and hence  $U(e; \hat{\theta})$  is supermodular in  $(e, \hat{\theta})$ .  $\square$

## B Methodological Detail - Text Embeddings

Text embeddings were computed for each post by inputting the post text into OpenAI’s `text-embedding-3-small` embedding model. The embedding model computes a 1,536 dimensional embedding vector for each post that encodes the meaning of the text in such a way that two posts with similar text will have a small distance between their embedding vectors. The raw embedding dimensions are learned from a large general purpose training corpus and are not tailored to the structure of our dataset. We therefore compute the principal components of the raw embedding vectors using our corpus. The principal components are orthogonal linear combinations of the original 1,536 embedding dimensions that explain the greatest variation across posts in our data. Figure 6 shows the scree plot of the principal components analysis, for the first 30 components. To keep our regression in (11) parsimonious, we decided to use only the first 10 components in the analysis. These components explain 23.7% of the total variation, but our results in Section 5 are robust to the specific choice of how many components to use. After running regression (11), we inspected the embedding dimensions that were most predictive of zaps. The first most predictive dimension appears to capture meta-posts (not necessarily in the `meta` territory) about Stacker News itself, such as tips for how to post and earn more sats. The second most predictive dimension appears to capture posts about geopolitical and macroeconomic events, such as a personal discussion about elections in Venezuela. The third most predictive

dimension appears to capture posts that ask questions about personal finance. We note that, although these embedding dimensions are the most strongly associated with zaps, other predictive aspects of the text content may not align with any single dimension and may instead be captured by combinations of dimensions. We therefore treat the embedding components primarily as control variables in the regression and avoid over-interpreting individual coefficients or assigning strong substantive meaning to any single dimension.

Figure 6: Scree Plot for Post Text Embeddings



*Note:* Shows the scree plot for post text embeddings, indicating the explained variance ratio for each principal component. The total explained variance at  $K = 10$  is 23.7%.